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A short review of relation extraction methods

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Abstract—On the basis of reviewing the history of relationship extraction, this paper divides the research into three categories: traditional machine learning based relation extraction method, deep learning-based relationship extraction method, and open domain relationship extraction method. Then, each method is divided into several different sub methods according to the characteristics of the processing. Based on the analysis of the relevant literature of each method, the relationship and difference between the various methods are briefly described.

Keywords- relation extraction; machine learning; Deep learning; Open relation extraction

I. INTRODUCTION

With the rapid development of big data, a large amount of information appears in front of people in the form of semi-structured or pure original text, Therefore, how to quickly and effectively extract the information users really need from massive information sources at low cost has become an urgent demand of contemporary computer research technology. In this context, information extraction (IE) technology emerges as the times require, Information extraction studies the method and technology of extracting specific entity, relationship and event from natural language text. Relation extraction (RE) is one of the key technologies of information extraction, the task of relation extraction can be described as: extracting entity1 and entity2 and their relationship types from text and representing them as a structure similar to (entity1, relation type, entity2). Relation extraction solves the problem of relationship classification between target entities in text, It is an important processing step to construct complex knowledge base system and knowledge map, such as machine translation, question answering system, text abstract, knowledge map and other intelligent applications, Therefore, with the rapid development of the research and application of knowledge mapping, relational extraction, as the core link of knowledge mapping construction, has gradually been widely concerned by the research community.

II. THE HISTORY AND RESEARCH STATUS OF RELATION EXTRACTION

1. history of relation extraction

Template extraction was first proposed in the seventh message understanding Conference (MUC) held by the Defense Advanced Research Projects Agency (DARPA) in 1998, and then developed into relational extraction. In this conference, the relationship extraction task was proposed as an evaluation task for the first time. The purpose of this task is to extract the location between entities by filling the relationship template slot_ of, Employee _ of and product_ the content of the evaluation corpus is mainly from the news related to the plane crash and space launch event in the New York Times, and the corresponding evaluation system is designed. MUC series of evaluation meetings have played a great role in promoting the research of relationship extraction [1].

In 1999, the National Institute of Standards and Technology (NIST) launched the automatic content extraction (ACE) to replace the previous MUC. This assessment supports automatic processing of language texts from various sources. The purpose of the study is to automatically extract the entities, relationships and events in the corpus, i.e. the entities, relationships and events in the corpus for identification and description [2]. ACE conference divided relationship extraction tasks into seven types in 2008; in 2009, ACE conference was formally included in the text analysis conference (TAC) organized by the National Institute of standards and technology, and the relationship extraction was included in the slot filling task in the knowledge base population field [3]. ACE conference provides the evaluation corpus of relation extraction, and constructs detailed entity relationship types, which further refines the relationship extraction task.

With the popularization of MUC and ACE, relational extraction technology has made great progress. Research hotspots have developed from the application of simple language model to the application of NLP technology using shallow parser or full parser, and the application of complex machine learning methods. The performance of relation extraction has been greatly improved.

2. research status of relation extraction

Driven by MUC and ACE evaluation meeting, many different methods related to relation extraction research have been produced. According to the development of relation extraction, this paper divides the methods of relation extraction into three categories, and then divides them into several different sub methods according to the characteristics of processing, and briefly describes the relations and differences between various methods. The following detailed classification methods.

A. Relation extraction method based on traditional machine learning

In 2000, Miller et al. [4] proposed a lexicalized Probabilistic Syntactic model, which was trained with the Pennsylvania tree database and achieved good results in the task of template relation extraction. The research of Miller et al. Shows that it is feasible to solve the problem of relation extraction by using machine learning method, and the performance of the system can be effectively improved by using lexical features and syntactic features to train the model. The method based on machine learning has gradually

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developed into a key idea to realize relation extraction. According to the different degree of human participation and dependence on the annotated corpus, the machine learning based relation extraction methods can be divided into three categories: guided machine learning methods, semi guided machine learning methods and unsupervised machine learning methods.

- Supervised machine learning

The supervised machine learning method regards relation extraction as a classification problem, which is to judge whether the entity pair belongs to the previously defined relation type through a series of characteristics of two entities. The main idea of using supervised learning method to solve the relationship extraction task is to train the model according to the existing manually labeled data, and then identify and extract the specific relationship. According to the different representation of relational instances, supervised machine learning methods can be divided into two categories: the method based on feature vector and the method based on kernel function.

The method based on feature vector firstly extracts a series of features from useful information such as part of speech and entity location from sentence context to train a classifier, thus completing the task of relation extraction. Kambhatla et al. [5] based on ACE RDC 2003 In order to reduce the dependence on semantic feature extraction tree, a maximum entropy model based relational classifier is constructed to reduce the dependence on semantic feature extraction tree. The results show that the rich linguistic features of entity context have rich value for relation representation, which lays the foundation for subsequent relation extraction. Giuliano et al. [6] used the evaluation materials of SEM eval-2007, selected the entity context information, distance and other features, and realized the relationship extraction task with the support vector machine (SVM) model, and achieved good results. Tratz et al. [7] extract features generated from a single word, including nouns and their context information, as well as a combination of nouns, and use a maximum entropy classifier trained with a large number of Boolean features to implement relation extraction tasks on the SemEval-2010 evaluation corpus. Che Wanxiang et al. [8] used the ACE RDC2004 corpus, selected the left and right words of each entity as features, and used Winnow and SVM machine learning algorithms to complete the automatic extraction of entity relationships, which also achieved good results. Chen Yu et al. [9] used ACE04 corpus and DBN (Deep Belief Nets) model for relation extraction tasks. Experimental results showed that character features are more suitable for Chinese relation extraction tasks than word features, and its effect is better than SVM and backpropagation networks.

Most learning algorithms rely on feature-based object representation. In other words, an object is transformed into a set of features $F_1,...,F_N$, thereby generating an N-dimensional vector. In many cases, data is not easy to express through features. For example, in most NLP problems, feature-based representation will produce an inherent local representation of the object, because it is computationally impossible to generate features that contain long-range dependencies. Therefore, researchers will replace feature-based methods based on kernel functions.

The method based on the kernel function refers to directly taking the structure tree as the processing object and using the kernel function to calculate the similarity between them to train the relationship classification model. How to design a kernel function for calculating the similarity of two entities is the most critical step. Zelenko et al. [10] used the existing shallow parsing system to generate the shallow syntactic analysis results of news articles, and trained support vector machines (SVM) and voting perceptions to extract people from the text-affiliation and organization-Positional relationship, where the experimental corpus includes 200 news articles from different news agencies and publications. Bunescu et al. [11] applied the shortest dependency path kernel function to complete the relation extraction task from the ACE corpus (NIST, 2000). It can be seen that the method based on the kernel function has strict matching constraints in the process of calculating the similarity, resulting in a low final recall rate. To solve this problem, the researchers introduced a convolution kernel function. The convolution kernel function calculates the similarity between discrete structures by counting the number of identical substructures. Qian et al. [12] used the composition dependency to maintain the path of the two entities, and at the same time removed the noise information from the syntactic parse tree, and finally formed a dynamic dependency tree, which further improved the efficiency of the convolution tree kernel function method. Zhang et al. [13] make full use of the excellent characteristics of the kernel function to explore the extraction of diversified knowledge relations and propose a method of modeling syntactic structure information based on the convolution kernel of parse tree to realize relation extraction. Research shows that the convolution tree kernel function effectively captures structured features and does not require extensive feature engineering.

- Semi-supervised machine learning method

Although the use of supervised machine learning methods to solve the relationship extraction problem can achieve good experimental results, due to the high dependence of supervised machine learning methods on the resources of the annotation corpus, it is difficult to adapt to the requirements of automatic processing. Relation extraction under the condition of manual participation and labeling of corpus resources has become a new hot spot in the research field, resulting in semi-supervised machine learning methods.

The semi-supervised relationship extraction method is to bootstrapping from the relationship seed. First, manually create a small number of relationship instances as the initial seed set. Secondly, use pattern learning or model training for continuous iterative learning, and finally generate relationship data Set and sequence patterns, this method reduces the dependence on manual annotation corpus to a certain extent. At the beginning, Brin [14] established the DIPRE system using the bootstrapping method. He selected the author and title pair from the World Wide Web as the seed entity relationship pair and extracted new examples

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from documents and sentences as labeled samples. And iterate, automatically create a new extraction template, the experiment used a repository of 24 million web pages, a total of 147G. Agichtein et al. [15] designed and implemented the Snowball relationship extraction method based on the DIPRE system. The main difference is that the Snowball system uses vectors to represent entities and tuples of relationships between entities and uses the similarity of vectors to iteratively increase the labeled samples. In each iteration, the model evaluates the quality of these patterns and tuples without manual intervention, and only retains the most reliable part to improve the overall quality. Chen Jinxiu et al. [16] used graph strategies to establish a graph-based semi-supervised extraction model, that is, an annotation transfer algorithm, which automatically recognizes the relationship between entities from unstructured text and improves the performance of relationship extraction.

Semi-supervised machine learning methods can effectively reduce manual participation and dependence on annotated corpus and can achieve large-scale text relation extraction tasks. It has been widely adopted. However, the bootstrap method has the phenomenon of Semantic Drift [17] in the iterative process, which affects the accuracy of the relationship extraction results. Therefore, how to obtain new relationship examples and extract templates with higher credibility is the focus of current research.

- Unsupervised machine learning method

The unsupervised machine learning method is a bottom-up extraction method, first extracting a large number of redundant corpus entities and relationships, and then labeling the cluster set. Some researchers use unsupervised machine learning methods to further reduce manual participation and dependence on labeled corpus.

Davidov et al. [18] used Web-based mining methods to automatically mine clustering patterns containing concept words and other words related to them. Yan et al. [19] used Wikipedia and web corpus to propose a clustering method that combines dependent features and shallow grammar templates and achieved good results. Hassan et al. [20] proposed an unsupervised relationship extraction method that relies on redundancy in large data sets and mutual enhancement based on graphs, which is used for automatic content extraction (ACE) relationship detection and feature description (RDC) tasks Relation extraction in. Zhang et al. [21] regarded relation extraction as a clustering problem based on a shallow syntactic tree, first modified the original tree core, and then used a top-down hierarchical clustering algorithm to group entity pairs into different clusters Finally, each cluster is marked by a demonstrative word, and unreliable clusters are pruned.

The advantage of the unsupervised method is that it does not rely on the current entity relationship type definition system. Therefore, it is feasible to transplant the algorithm to different areas. The disadvantage is that a large-scale corpus is generally required as support. The final clustering results have high requirements on the quality of the corpus, the relationship labeling is relatively wide, and the relationship name is difficult to accurately describe, so the accuracy and recall rate of relationship recognition are generally not high.

B. Relation extraction method based on deep learning

Because the feature vectors selected by the traditional machine learning relationship extraction method are over-reliant on manual completion, a large amount of domain expertise is also required, while the relationship extraction method of deep learning automatically obtains the model through training a large amount of data, without the need for manual feature extraction. Zeng et al. [22] Used deep neural network (DNN) for the first time to extract vocabulary level features and sentence level features, first convert all word tags into vectors as input, and then extract vocabulary level features based on given nouns. At the same time, the convolution method is used to learn sentence-level features, and the two-level features are connected in series to form the final feature vector, and finally the relationship is extracted through the softmax layer. This method effectively reduces the excessive dependence on manual performance of the feature vectors selected by traditional relationship extraction methods. Socher et al. [23] proposed the use of recurrent neural networks (RNN) to solve relation extraction tasks. First, the sentence is syntactically parsed, and then the vector representation is learned for each node on the syntax tree. The recurrent neural network starts from the word vector at the top of the syntax tree and merges iteratively according to the syntax structure of the sentence. Finally, the vector representation of the sentence is obtained. And used for relationship classification. Although this method can effectively consider the syntactic structure information of the sentence, it cannot consider the position of the two entities in the sentence and the sentence meaning information. Santos et al. [24] proposed a new convolutional neural network for relation extraction, which uses a new loss function of pairwise sorting, which can easily reduce the influence of artificial classes. Miwa et al. [25] Proposed a new relationship extraction model based on end-to-end neural network. The model uses bidirectional LSTM and tree LSTM to simultaneously model entities and sentences. Lin et al. [26] proposed a neural network model based on a sentence-level attention mechanism, which uses convolutional neural networks to embed the semantics of sentences, and then constructs sentence-level attention on multiple instances, which can make effective sentences through continuous learning Get higher weight, while noisy sentences get less weight. This model can make full use of all information sentences and effectively reduce the impact of incorrectly marked instances.

C. Open relation extraction

Since traditional relation extraction is based on specific domains and specific relationships, the task of relation extraction is time-consuming, laborious, and costly, which is not conducive to the expansion of corpus types. In recent years, the method of extracting entity relationships in open domains has attracted more and more attention. Open-domain relationship extraction does not limit the relationship category and target text and has incomparable advantages in cross-domain and later expansion. In addition, this method does not need to manually formulate relationship types in advance, reducing the burden of manual labeling. The system designed in this way has strong portability and

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promotes the development of relation extraction. Yao et al. [27] Made full use of the advantages of open information extraction and proposed a hierarchical clustering relationship extraction model based on LDA. Compared with traditional relationship extraction methods, this model has faster calculation speed and higher accuracy. Yao Xianming et al. [28] proposed a method for extracting multiple entity relationships in the Chinese field. From the perspective of dependent grammar, this method extracts multiple entity relationships represented by the subject-predicate-object structure by analyzing the results of syntactic analysis and has achieved good results.

Although the current open field relationship extraction has achieved good results, there are still some difficulties. How to continue to improve the accuracy and recall rate of entity binary relationships, further realize the extraction of multiple relationships between entities, and improve the relationship between entities. The effective use of information and the development of unified evaluation standards.

III. EVALUATION INDEX FOR RELATION EXTRACTION

The evaluation criteria for the final effect of the relationship extraction task generally include three common evaluation indicators: precision, recall, and F1 value.

The precision is for the prediction result, and its meaning is the probability that the sample is positive among all the samples that are predicted to be positive, the calculation formula is

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

Among them, TP represents the number of positive examples that are correctly predicted, the number of samples that are actually positive and are predicted by the model as positive; FP is the number of positive examples that are incorrectly predicted, which is actually negative but is predicted as positive by the model The number of samples.

Recall rate is for the original sample, and its meaning is the probability of being predicted as a positive sample in a sample that is actually positive, the calculation formula is

$$R = \frac{TP}{TP + FN}$$

Where FN is expressed as the number of falsely predicted negative examples, the actual number of positive examples but the number of negative examples predicted by the model.

We hope that the accuracy and recall rate of the experimental results are high, but in fact they are contradictory and cannot be double high. But usually we can define a new indicator based on the balance point between them: F1 score (F1-Score). The F1 score considers both the accuracy rate and the recall rate, so that the two can reach the highest at the same time and strike a balance, the F1 score expression is

$$F1 = \frac{2 \times P \times R}{P + R}$$

IV. CONCLUSIONS

In short, relation extraction is one of the important research topics in natural language processing. After years of development, the related theories and methods of relation extraction have become more and more perfect. The research content has changed from the initial limited field and limited type of relationship extraction to the automatic extraction of entity relationships for the open field. Relation extraction has important application prospects as a basic support technology in intelligent question answering, knowledge graphs, automatic retrieval and other fields, and the extraction effect still has a lot of room for improvement.

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